

RECUBRIMIENTO MINIMAL

④ $R = \{A \dots F\}$ $\Phi = \{EF \rightarrow D, E \rightarrow A, AB \rightarrow C, F \rightarrow B, EAD \rightarrow B, BE \rightarrow D, A \rightarrow E\}$

1) $\Phi_1 = \{EF \rightarrow D, E \rightarrow A, AB \rightarrow C, F \rightarrow B, \overset{+}{EAD} \rightarrow B, BE \rightarrow D, BE \rightarrow F, A \rightarrow E\}$

2) Cierres respecto a Φ_1

1) $E \notin F^+ = \{F, B\}$ $F \notin E^+ = \{E, A\}$

2) $A \notin B^+ = \{B\}$ $B \notin A^+ = \{A, E\}$

3) $E \in AD^+ = \{A, D, E, B, F, C\}$ $A \notin D^+ = \{D\}$ $D \notin A^+ = \{A, E\}$

4) $B \notin E^+ = \{E, A\}$ $E \notin B^+ = \{B\}$

$\Phi_2 = \{EF \rightarrow D, E \rightarrow A, AB \rightarrow C, F \rightarrow B, AD \rightarrow B, BE \rightarrow D, BE \rightarrow F, A \rightarrow E\}$

3) Cierres respecto a Φ_2 - $\Phi_2(i)$ $i=1 \dots$

1) $D \in EF^+ = \{E, F, A, B, C, D\}$

5) $B \notin AD^+ = \{A, D, E\}$

2) $A \notin E^+ = \{E\}$

6) $D \notin BE^+ = \{B, E, F, A, C\}$

3) $C \notin AB^+ = \{A, B, E, D, F\}$

7) $F \notin BE^+ = \{B, E, D, A, C\}$

4) $B \notin F^+ = \{F\}$

8) $E \notin A^+ = \{A\}$

$\Phi_3 = \{E \rightarrow A, AB \rightarrow C, F \rightarrow B, AD \rightarrow B, BE \rightarrow D, A \rightarrow E\}$

⑦ $R = \{A \dots G\}$ $\Phi = \{BG \rightarrow F, D \rightarrow E, FG \rightarrow BE, D \rightarrow B, E \rightarrow B, EF \rightarrow G, BDG \rightarrow A, B \rightarrow AE\}$

1) $\Phi_1 = \{BG \rightarrow F, D \rightarrow E, FG \rightarrow B, FG \rightarrow E, D \rightarrow B, E \rightarrow B, EF \rightarrow G, BDG \rightarrow A, B \rightarrow A, B \rightarrow E\}$

2) Elementos varios

$B \notin G^+ = \{G\}$ $G \notin B^+ = \{B, A, E\}$ $E \notin F^+ = \{F\}$ $F \notin E^+ = \{E, B, A\}$

$F \in EG^+ = \{G\}$ $G \notin F^+ = \{F\}$

$B \in DG^+ = \{D, G, E, B, A, F\}$ $D \notin G^+ = \{G\}$ $G \notin D^+ = \{D, B, A, E\}$

$\Phi_2 = \{BG \rightarrow F, D \rightarrow E, FG \rightarrow B, FG \rightarrow E, D \rightarrow B, E \rightarrow B, EF \rightarrow G, DG \rightarrow A, B \rightarrow A, B \rightarrow E\}$

3) dependencias redundantes:

$F \notin BG^+ = \{B, G, A, E, B\}$ $E \in D^+ = \{D, B, A, E\}$

$B \in FG^+ = \{F, G, E, B, A\}$ $E \notin FG^+ = \{F, G\}$

$B \notin D^+ = \{D\}$ $B \notin E^+ = \{E\}$ $G \notin EF^+ = \{E, F, B, A\}$

$A \in DG^+ = \{D, G, B, A, E, F\}$ $A \notin B^+ = \{B, E\}$ $E \notin B^+ = \{B\}$

$\Phi_3 = \{BG \rightarrow F, FG \rightarrow E, D \rightarrow B, E \rightarrow B, EF \rightarrow G, B \rightarrow A, B \rightarrow E\}$

HALLAR CC

① $R = \{A \dots G\}$

$DFS = \{GB \rightarrow F, D \rightarrow E, GF \rightarrow E, D \rightarrow B, E \rightarrow B, EF \rightarrow G, GD \rightarrow A, B \rightarrow E, GF \rightarrow B, B \rightarrow A\}$

1) $CD^+ = \{C, D, E\} \neq R$

Resto	Ambos lados	Solo Derecha
C, D	B, E, F, G	A

2) $CDB^+ = \{C, D, B, E, A\} \neq R$

$CDE^+ = \{C, D, E, B, A\} \neq R$

$CDG^+ = \{C, D, F, B, E, A, G\} = R$

$CDG^+ = \{C, D, G, B, A, E, F\} = R$

3) $CDBE^+ = \{C, D, B, E, A\} \neq R$

$CDBF$ NO
 $CDBG$ NO

$CDEB^+ \neq R$
 $CDEF$ NO
 $CDEG$ NO

4) $CDBEF$ NO
 $CDBEG$ NO

5) $CK = \{CDF, CDG\}$

⑤ $R = \{A \dots G\}$

$DFS = \{AB \rightarrow C, F \rightarrow B, EF \rightarrow G, E \rightarrow A, AG \rightarrow B, BE \rightarrow F, A \rightarrow E\}$

Resto	Izda. deha.	Solo deha.
D	A, B, E, F, G	C

1) $D^+ = \{D\} \neq R$

2) $DA^+ = \{D, A, E\} \neq R$

$DB^+ = \{D, B\} \neq R$

$DE^+ = \{D, E, A\} \neq R$

$DF^+ = \{D, F, B\} \neq R$

$DG^+ = \{D, G\} \neq R$

4)

$DAEB$ NO

$DAEF$ NO

$DAEG$ NO

$DBFA$ NO

$DBFE$ NO

$DBG^+ = \{D, B, F, G\} \neq R$

DBG NO

$DBGE$ NO

* $DBG^+ = \{D, B, G, F\} \neq R$

$DFGA$ NO

$DFGB^+ = \{D, B, G, F\} \neq R$

$DFGE$ NO

5) $DEGFA$ NO
 $DEGFE$ NO

3)

$DAE^+ = \{D, A, E, B, C, F, G\} = R$

* $DAE^+ = \{D, A, E\} \neq R$

$DAF^+ = \{D, A, F, E, G, B, C\} = R$

$DAG^+ = \{D, A, G, B, C, E, F\} = R$

$DBE^+ = \{D, B, E, A, C, F, G\} = R$

* $DBF^+ = \{D, B, F\} \neq R$

* $DBG^+ = \{D, B, G\} \neq R$

$DEF^+ = \{D, E, F, A, G, B, C\} = R$

$DEG^+ = \{D, E, G, A, B, C, F\} = R$

* $DFG^+ = \{D, F, G, B\} \neq R$

6) $CK = \{ABD, ADF, ADG, BDE, DEF, DEG\}$

FNBC

② $R = \{A..G\}$ $\Phi = \{AB \rightarrow C, F \rightarrow B, EF \rightarrow G, AG \rightarrow B, BE \rightarrow F\}$

- CC:

resto	$\rightarrow$ da delta.	Solo delta.
A, D, E	B, F, G	C
$CK = \{AEDB, ADEF, ADEG\}$		

1) $ADE^+ = \{A, D, E\} \neq R$

2) $ADEB^+ = \{A, D, E, B, C, F, G\} = R$
 $ADEF^+ = \{A, D, E, F, G, B, C\} = R$
 $ADEG^+ = \{A, D, E, G, B, F, C\} = R$

- FNBC:

1)

$R_1 = \{A, B, D, E, F, G\}$ $\Phi_1 = \{F \rightarrow B, EF \rightarrow G, AG \rightarrow B, BE \rightarrow F\}$
 $CK_1 = CK$

$R_2 = \{A, B, C\}$ $\Phi_2 = \{AB \rightarrow C\}$ $CK_2 = \{AB\}$ ✓

2)

$R_{1.1} = \{A, B, D, E, F\}$ $\Phi_{1.1} = \{F \rightarrow B, BE \rightarrow F\}$
 $CK_{1.1} = \{AEDB, ADEF\}$

$R_{1.2} = \{E, F, G\}$ $\Phi_{1.2} = \{EF \rightarrow G\}$ $CK_{1.2} = \{EF\}$ ✓

3)

$R_{1.1.1} = \{A, D, E, F\}$ $\Phi_{1.1.1} = \{\emptyset\}$ $CK_{1.1.2} = \{ADEF\}$ ✓
 $R_{1.1.2} = \{B, F\}$ $\Phi_{1.1.2} = \{F \rightarrow B\}$ $CK_{1.1.2} = \{B\}$ ✓

Sol: $\{(R_2, \Phi_2), (R_{1.2}, \Phi_{1.2}), (R_{1.1.1}, \Phi_{1.1.1}), (R_{1.1.2}, \Phi_{1.1.2})\}$